

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Double-blind Study to Evaluate Pembrolizumab Plus Chemotherapy vs Placebo Plus Chemotherapy as Neoadjuvant Therapy and Pembrolizumab vs Placebo as Adjuvant Therapy for Triple Negative Breast Cancer (TNBC)

Short Title/Acronym: KEYNOTE-522 **Protocol Number:** MK-3475-522 **NCT Number:** NCT03036488 **Posting ID:** 851420532061182204 **Sponsor/Company Name:** Merck Sharp & Dohme Corp. (Merck) **Investigational Product:** Pembrolizumab (KEYTRUDA®) **Preferred UMLS Name:** Pembrolizumab therapy for triple negative breast cancer **Phase of Development:** Phase III **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of pembrolizumab (MK-3475) plus chemotherapy versus placebo plus chemotherapy as a neoadjuvant therapy, followed by pembrolizumab versus placebo as an adjuvant therapy in participants who have triple negative breast cancer (TNBC). The primary hypothesis is that pembrolizumab is superior to placebo, in combination with chemotherapy, as measured by the rate of Pathological Complete Response (pCR) and/or Event-Free Survival (EFS). **Secondary Objectives:** To evaluate pCR rate by different definitions (e.g., ypT0 ypN0 vs ypT0/Tis ypN0), and Event-Free Survival (EFS) in participants with tumors expressing PD-L1.

2.2 Background & Rationale Given the high rates of recurrence within the first 5 years of diagnosis, patients with high-risk early-stage TNBC need new treatment options. Anti-PD-1 therapy combined with neoadjuvant chemotherapy has demonstrated the potential to significantly increase the pathological complete response (pCR) rate and improve event-free survival in this patient population, addressing a critical unmet medical need.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 2:1 (Pembrolizumab+Chemotherapy : Placebo+Chemotherapy)

3.2 Planned Enrollment & Duration Target **SN\$:** 1,174 evaluable patients **Number of Sites:** Multicenter (Global) **Estimated Study Start:** Mar-2017 **Estimated Primary Completion / Trial Status:** COMPLETED. Status Description: Study has been completed.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Has newly diagnosed, locally advanced, centrally confirmed TNBC, defined as previously untreated locally advanced non-metastatic (M0) TNBC with the following combined primary tumor (T) and regional lymph node (N) staging per AJCC criteria: T1c, N1-N2; T2, N0-N2; T3, N0-N2; or T4a-d, N0-N2. 3. Provides a core needle biopsy consisting of at least 2 separate tumor cores from the primary tumor at screening. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 performed within 10 days of treatment initiation. 5. Demonstrates adequate organ function. 6. Willing to use an adequate method of contraception for the course of the study through 12 months after the last dose of study treatment for participants receiving cyclophosphamide, and 6 months for participants who did not.

4.2 Exclusion Criteria 1. History of invasive malignancy $\leq$ 5 years prior to signing informed consent (except adequately treated basal or squamous cell skin cancer, or in situ cervical cancer). 2. Received prior chemotherapy, targeted therapy, and radiation therapy within the past 12 months. 3. Prior therapy with an anti-PD-1, anti-PD-L1, anti-PD-L2 agent or an agent directed to another co-inhibitory T-cell receptor (e.g., CTLA-4, OX-40, CD137) or previous participation in a pembrolizumab study. 4. Participating in or has participated in an interventional clinical study within 4 weeks of the first dose. 5. Received a live vaccine within 30 days of the first dose. 6. Active autoimmune disease requiring systemic treatment in the past 2 years. 7. Diagnosis of immunodeficiency or receiving systemic steroid therapy (> 10 mg daily of prednisone or equivalent) or other immunosuppressive therapy within 7 days prior to first dose. 8. Known history of HIV, active Hepatitis B or Hepatitis C, or active tuberculosis (TB). 9. History of (non-infectious) pneumonitis that required steroids or current pneumonitis. 10. Active infection requiring systemic therapy. 11. Significant cardiovascular disease (e.g., MI, ACS, bypass/stenting within the last 6 months, CHF NYHA Class II-IV, or history of CHF NYHA Class III-IV). 12. Pregnant, breastfeeding, or expecting to conceive children within the projected study duration. 13. Known hypersensitivity to the components of the study treatment or its analogs.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Neoadjuvant Phase (Approx. 24 weeks / 8 cycles):** Pembrolizumab 200 mg every 3 weeks (Q3W) plus chemotherapy (paclitaxel + carboplatin for 4 cycles, followed by cyclophosphamide [ATC: L01AA01; Trade Names: Cytosan, Endoxana] + either doxorubicin or epirubicin for 4 cycles). * **Surgery:** Definitive surgery occurs 3-6 weeks after conclusion of

the last cycle of neoadjuvant treatment. * **Adjuvant Phase (Approx. 27 weeks / 9 cycles):** Following definitive surgery, pembrolizumab 200 mg Q3W. **Route of Administration:** Intravenous (IV) infusion.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for symptom management and chemotherapy-related prophylaxis. **Prohibited:** Other investigational agents, live-virus vaccines (within 30 days of first dose), and systemic immunosuppressive therapies (>10mg prednisone equivalent within 7 days of dose).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy (2 cores), Baseline Imaging, ECOG
Neoadjuvant Phase	Weeks 1-24	Randomization, First Dose, IV Infusions (Pembrolizumab + Chemo), Safety Labs
Surgery	3-6 weeks post-neoadjuvant	Definitive breast surgery, pCR evaluation
Adjuvant Phase	Weeks 27-54	IV Infusions (Pembrolizumab monotherapy), Efficacy Assessment, PROs
Follow-up	Long-term	Event-free survival tracking, OS tracking, safety monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. Pathological Complete Response (pCR)

Rate: Defined as pathological stage ypT0/Tis ypN0 (i.e., no invasive residual in breast or nodes; noninvasive breast residuals allowed) assessed by the local pathologist at the time of definitive surgery. 2. **Event-Free Survival (EFS):** Defined as the time from randomization to the first occurrence of disease progression that precludes surgery, a local/distant recurrence, a second primary malignancy, or death from any cause as assessed by Investigator.

7.2 Secondary Endpoints 1. pCR rate using an alternative definition, ypT0 ypN0 (i.e., no invasive or noninvasive residual in breast or nodes) at the time of definitive surgery. 2. pCR rate using the definition of ypT0/Tis ypN0 at the time of definitive surgery. 3. Event-Free Survival (EFS) in participants with tumors expressing PD-L1.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical assessments, laboratory evaluations, and patient reporting, with a specific focus on Immune-Mediated Adverse Events (IMAEs).

Serious Adverse Events (SAEs): Reported within 24 hours of site

awareness. **Data Monitoring Committee (DMC):** External independent data monitoring committee.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy endpoints (pCR, EFS). Safety Set for all AE/SAE evaluations.

Sample Size Justification: $N=1,174$ appropriately powered ($\alpha = 0.05$) to detect statistically significant improvements in primary endpoints. **Handling of Missing Data:** Standard oncology imputation methods; Kaplan-Meier methodology for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard onsite and centralized data monitoring.

Audit Trail: eCRF locking and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-3475-522 **Registry Number:** NCT03036488 **Posting ID:** 851420532061182204 **Sponsor/Lead Organization:** Merck **Study Title:** Study of Pembrolizumab (MK-3475) Plus Chemotherapy vs Placebo Plus Chemotherapy as Neoadjuvant Therapy and Pembrolizumab vs Placebo as Adjuvant Therapy in Participants With Triple Negative Breast Cancer (TNBC) (MK-3475-522/KEYNOTE-522) **Trial Status:** COMPLETED

12. Clinical Framework (Oncology Specific)

Primary Disease Area / Condition: Triple Negative Breast Neoplasms **Semantic Type:** Neoplastic Process **Condition Definition:** Breast neoplasms that do not express ESTROGEN RECEPTORS; PROGESTERONE RECEPTORS; and do not overexpress the NEU RECEPTOR/HER-2 PROTO-ONCOGENE PROTEIN. **MeSH Code:** D064726 **SNOMED ID:** 709088003 **Medical Methodology:** Neoadjuvant Chemo-Immunotherapy followed by Adjuvant Immunotherapy **Optimization Target:** Improvement of pCR and reduction in the risk of EFS events

13. Study Procedures & Methodology

Management Pathway: Pre-operative chemo-immunotherapy followed by definitive surgery, subsequently followed by post-operative immunotherapy **Education Protocol (HCP):** Management of Immune-

Mediated Adverse Events (IMAEs) **Education Protocol (Patient):**
Patient-reported outcome (PRO) questionnaire completion **Quality**
Audit Mechanism: Independent central review of imaging and
pathology where applicable.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up for disease recurrence and survival.
Onsite Visit Cadence: Every 3 weeks (Q3W) during the treatment
phases. **Remote Monitoring Frequency:** Routine survival follow-up
post-treatment. **Checklist Review Interval:** Clinical and
radiographic evaluations at regular intervals during follow-up.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				